Depth and Shape Inference (IV)

Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

Stereopsis

41
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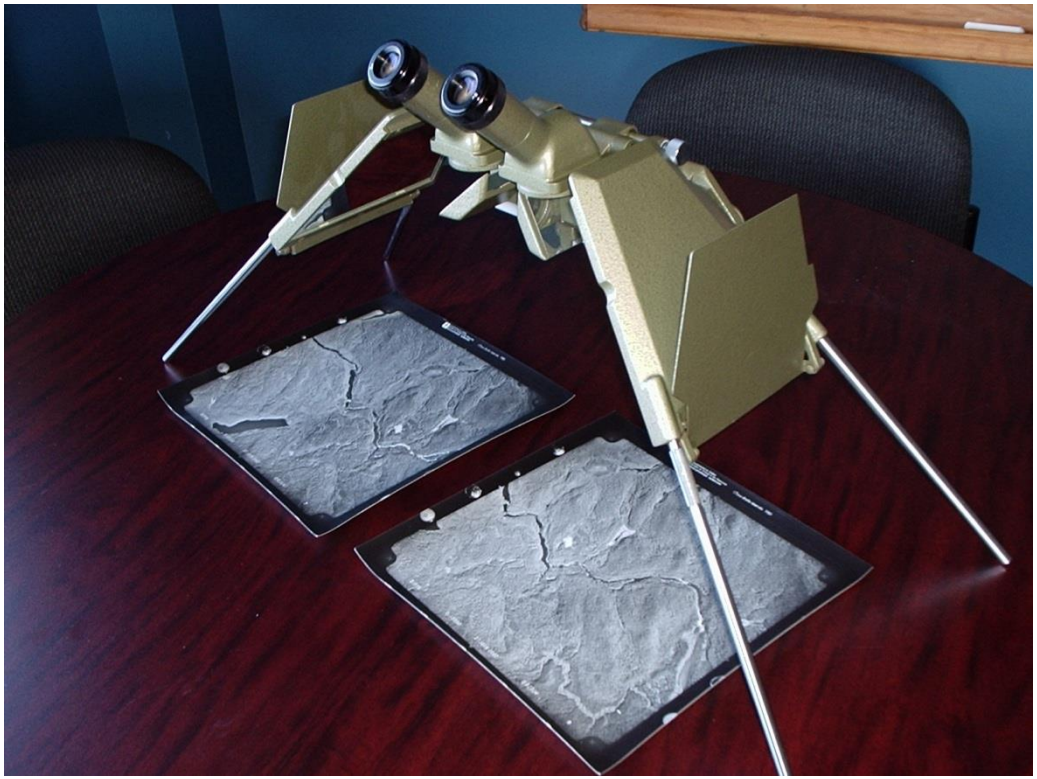
Stereopsis



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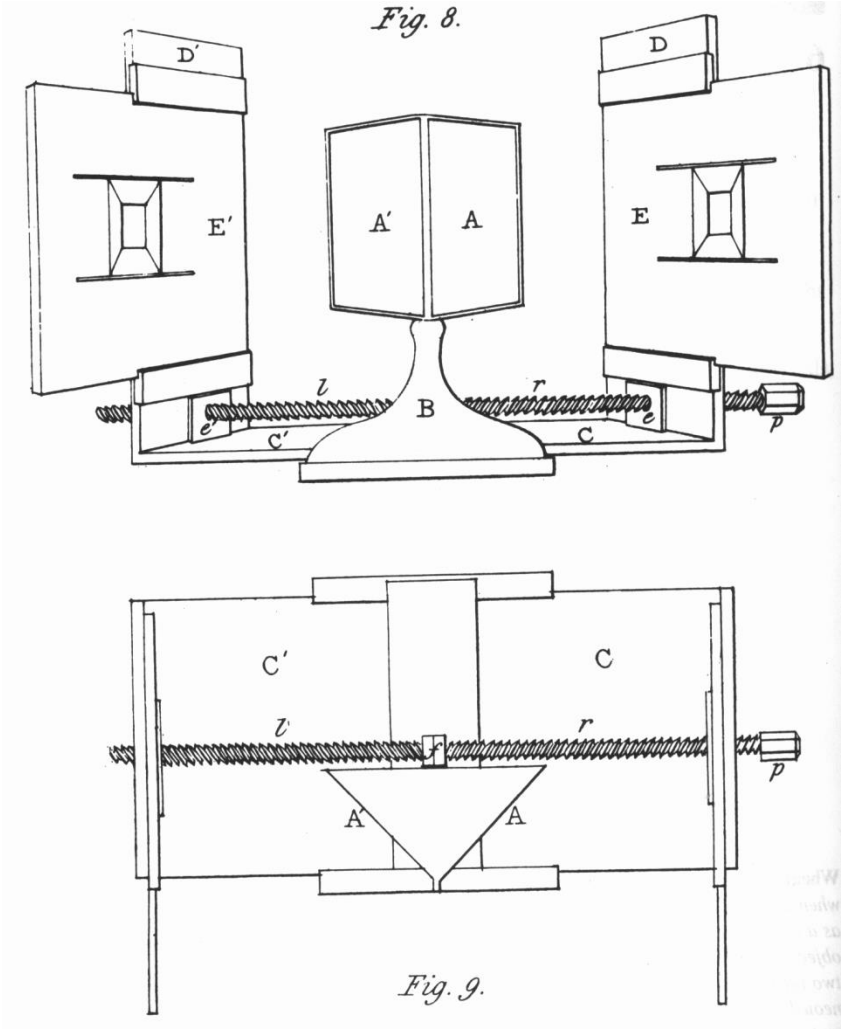

Wheatstone Mirror Stereoscope, 1838

Stereopsis

Stereopsis

Where does depth emerge?

"…if the two [blind man's] hands ***f*** and ***g*** each hold a stick, ***i*** and ***h***, with which they touch object ***K***, although the soul is otherwise ignorant of the length of these sticks, nevertheless because it knows the distance between the two points ***f*** and ***g*** and the size of the angles ***fgh*** and ***gfi***, it will be able to know, as if through natural geometry, where object ***K*** is"

[Descartes, 1664]

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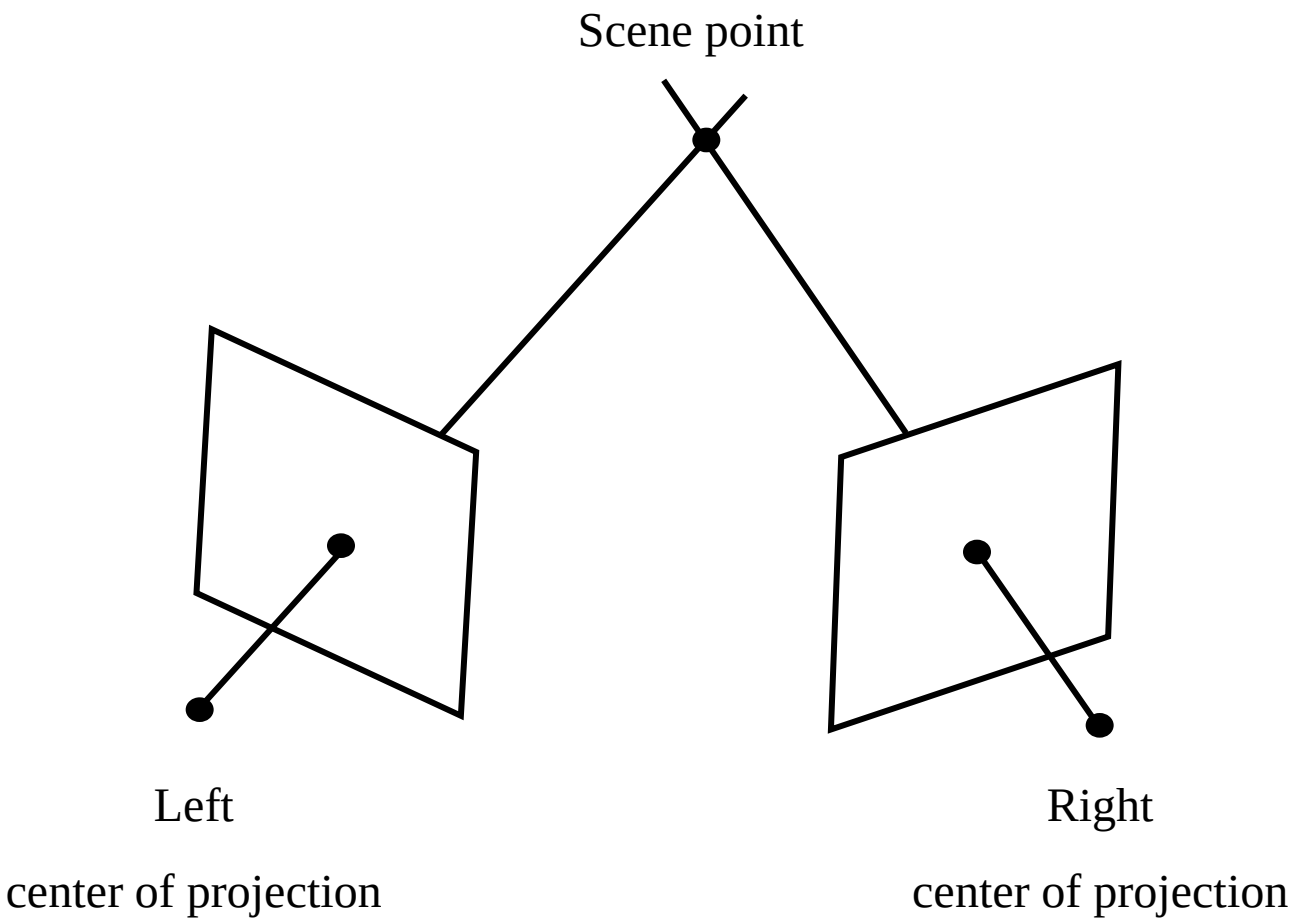
Depth via triangulation



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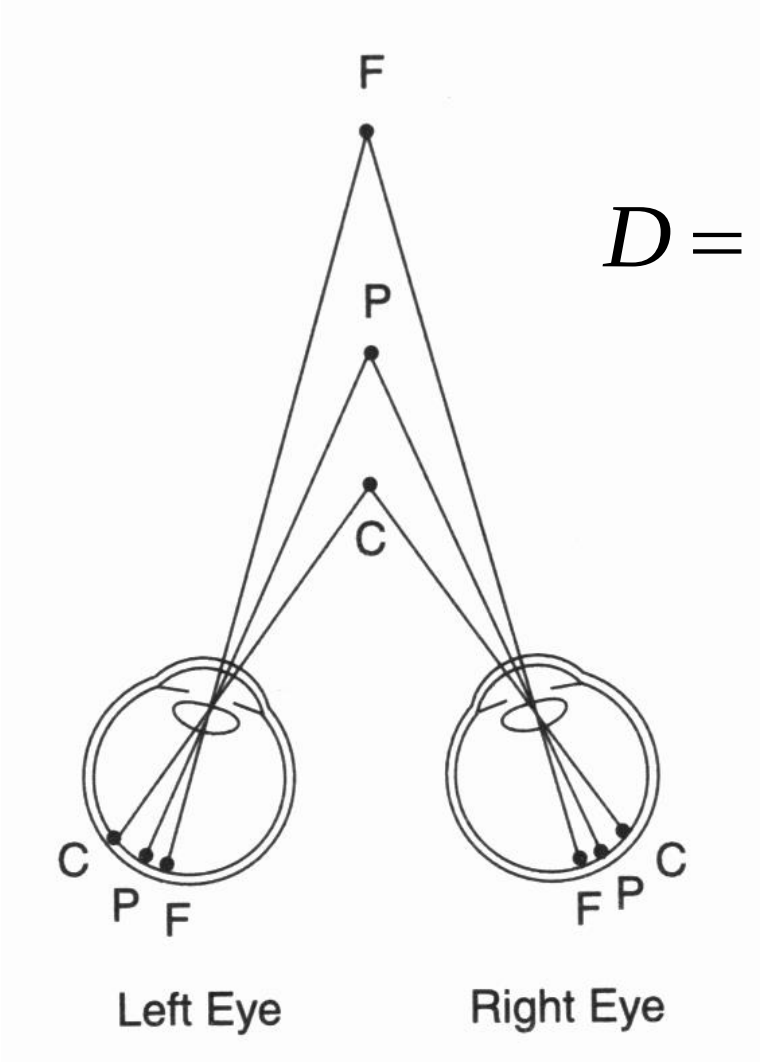
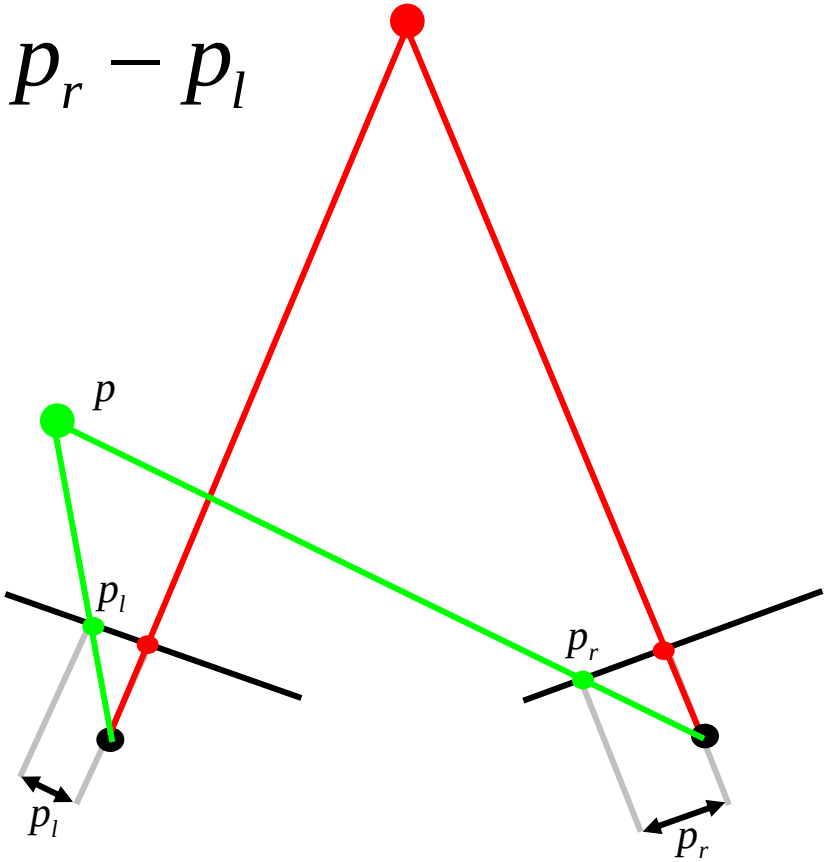
Binocular disparity



$$D = \Delta p = p_r - p_l$$



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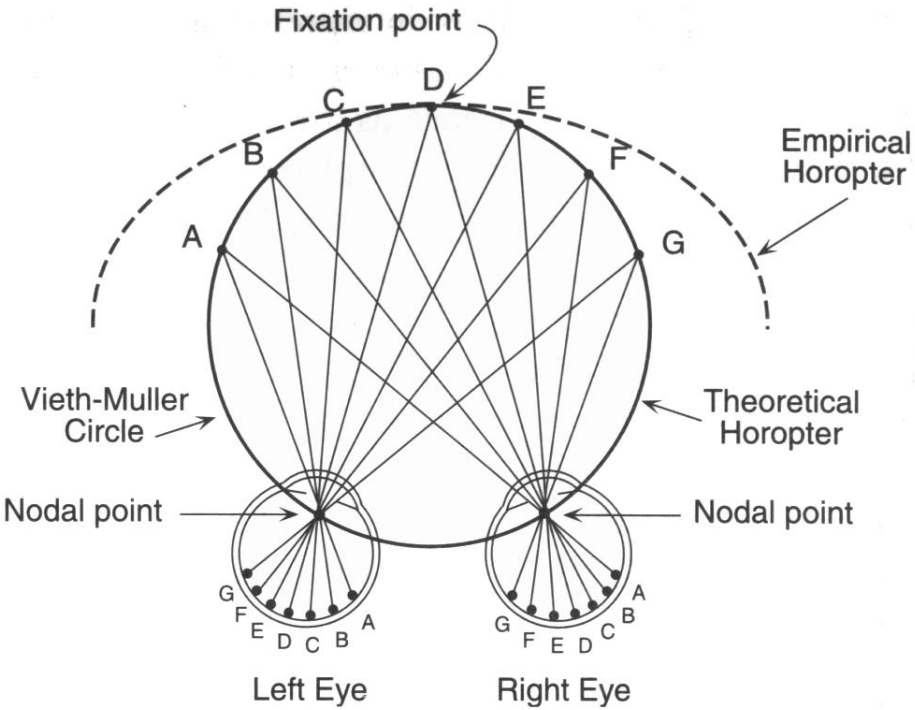
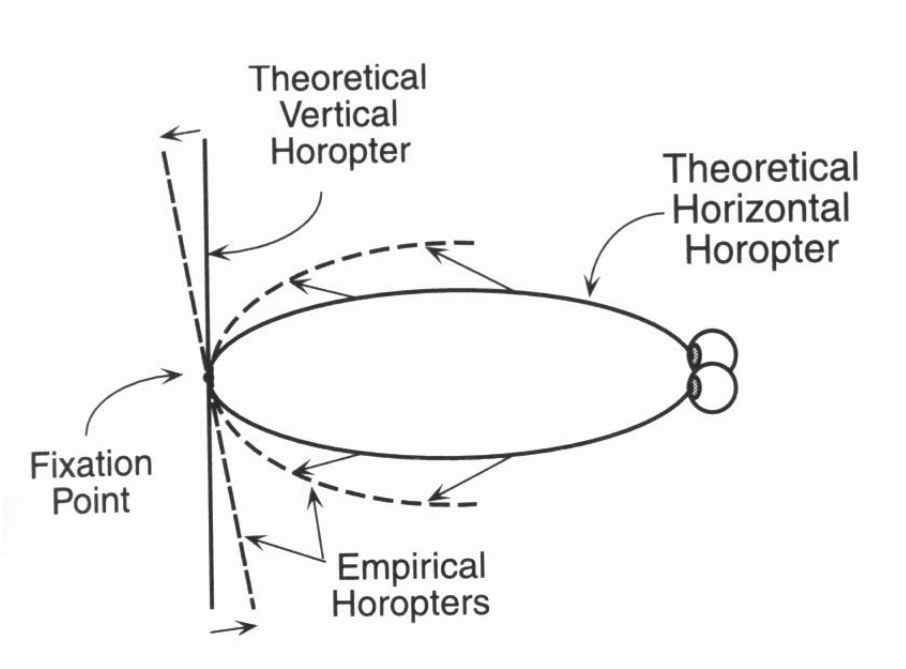
The Horopter (theoretical and empirical)





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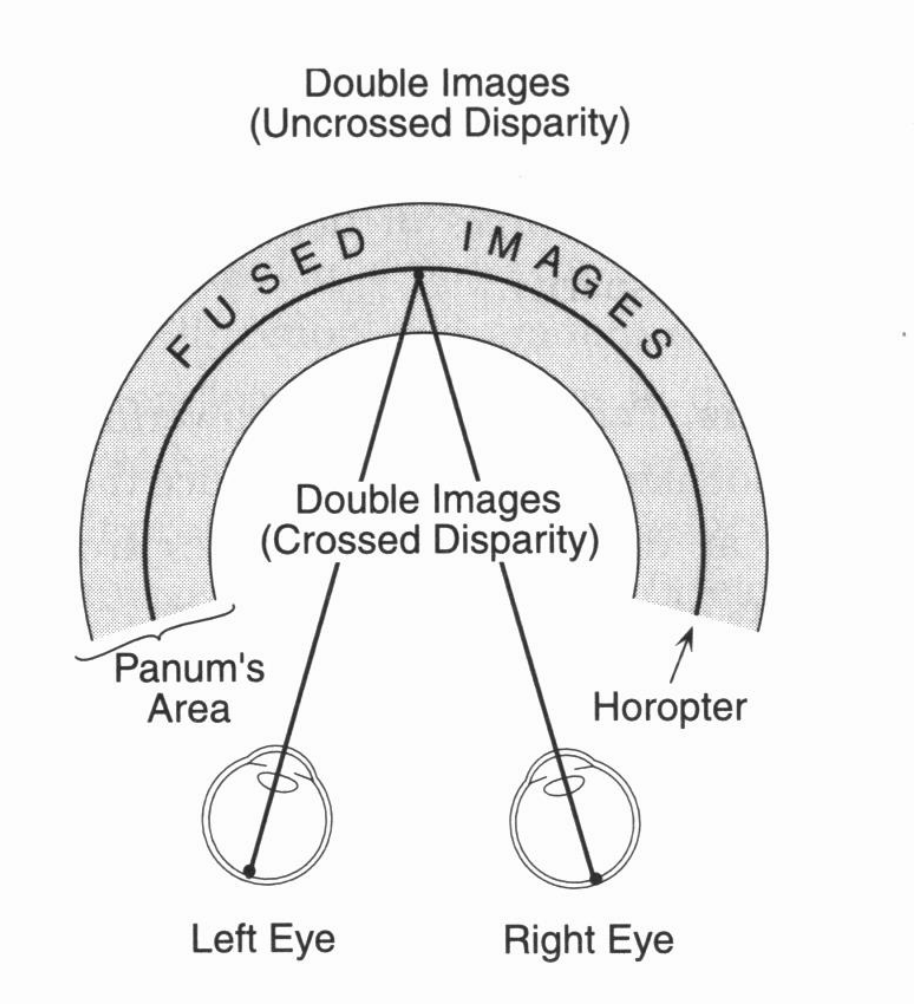
Panum's Area



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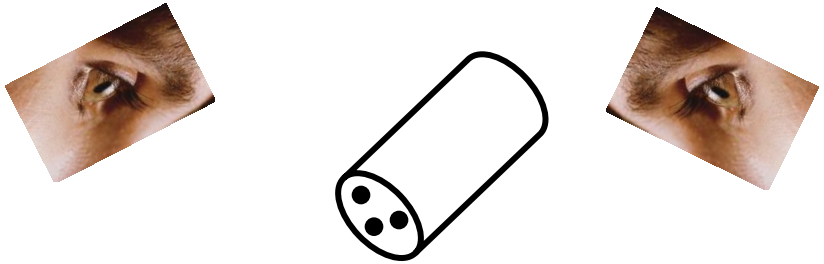
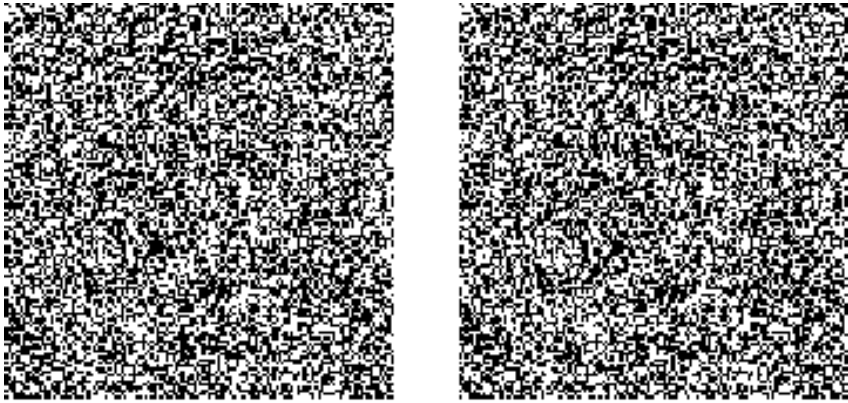
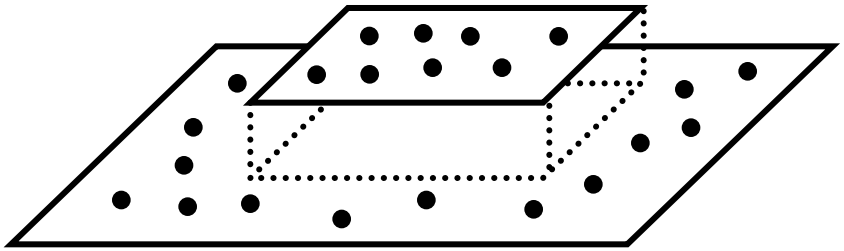
Random dot stereograms







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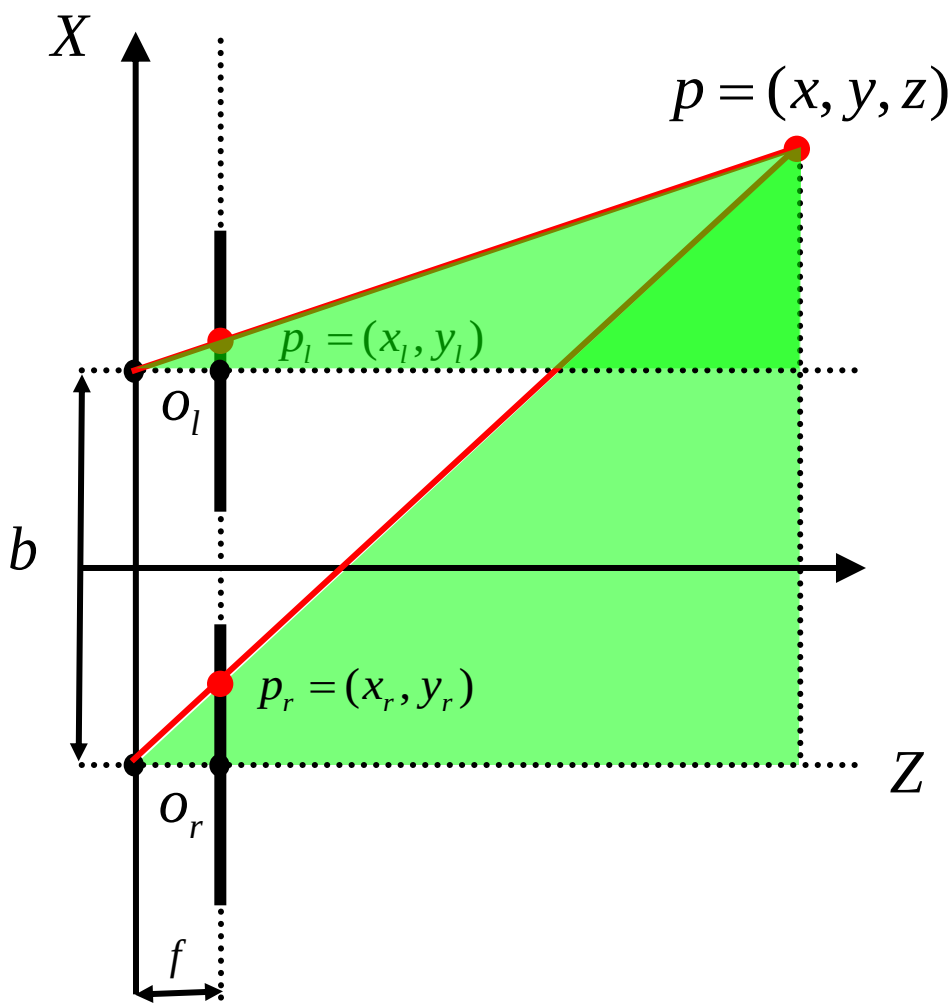
Depth reconstruction – Rectified Calibrated Stereo



$$\frac{x + \frac{b}{2}}{z} = \frac{x_r}{f}$$

$$\frac{x - \frac{b}{2}}{z} = \frac{x_l}{f}$$

$$\boxed{z = \frac{bf}{x_r - x_l} = \frac{bf}{D}}$$

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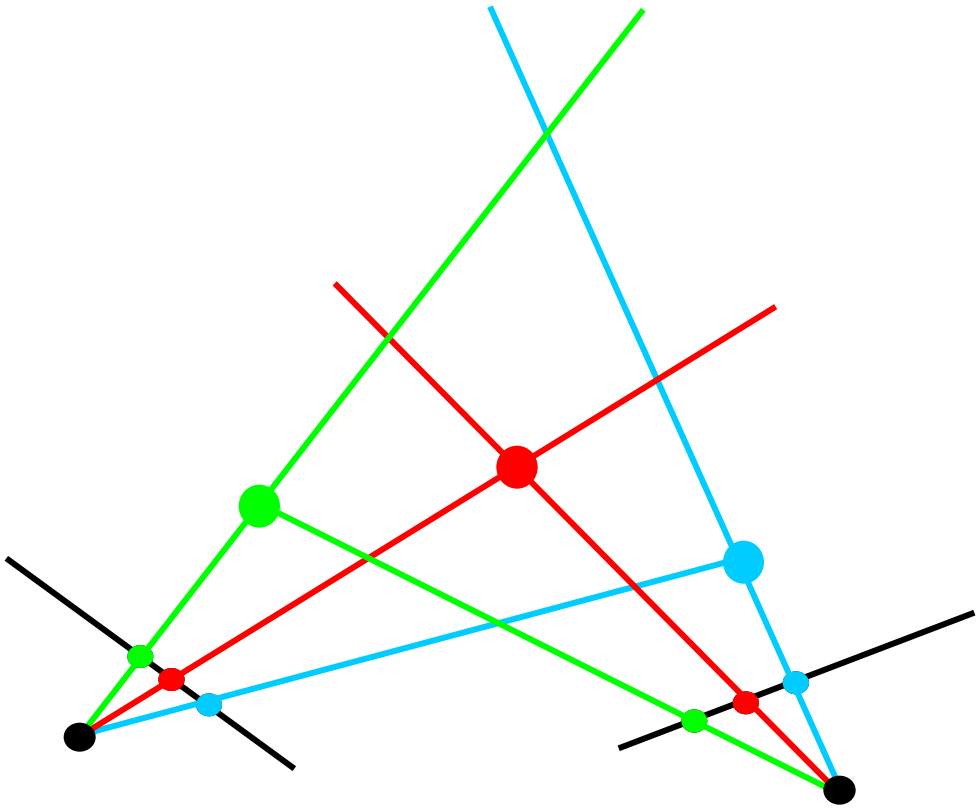
Stereo is solved?



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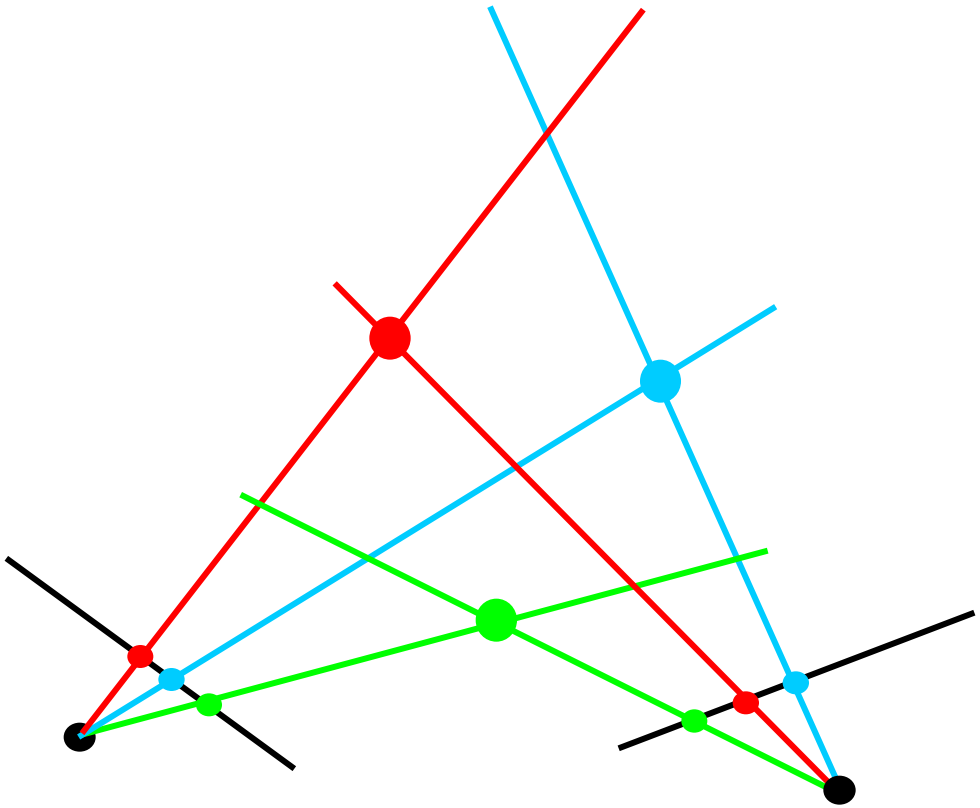
Stereo is solved?



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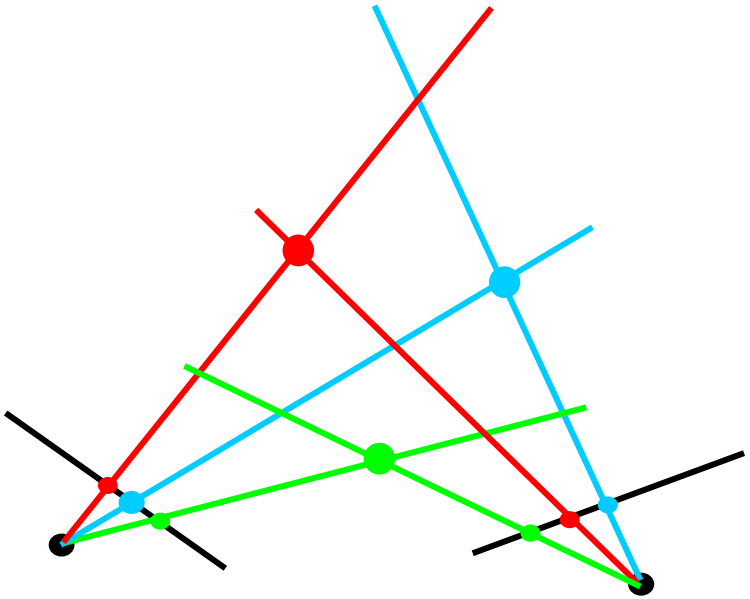
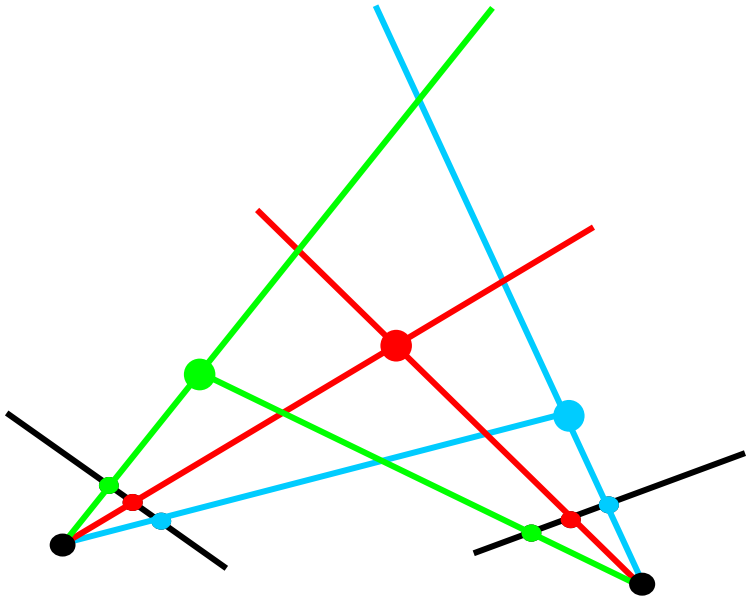
The correspondence problem





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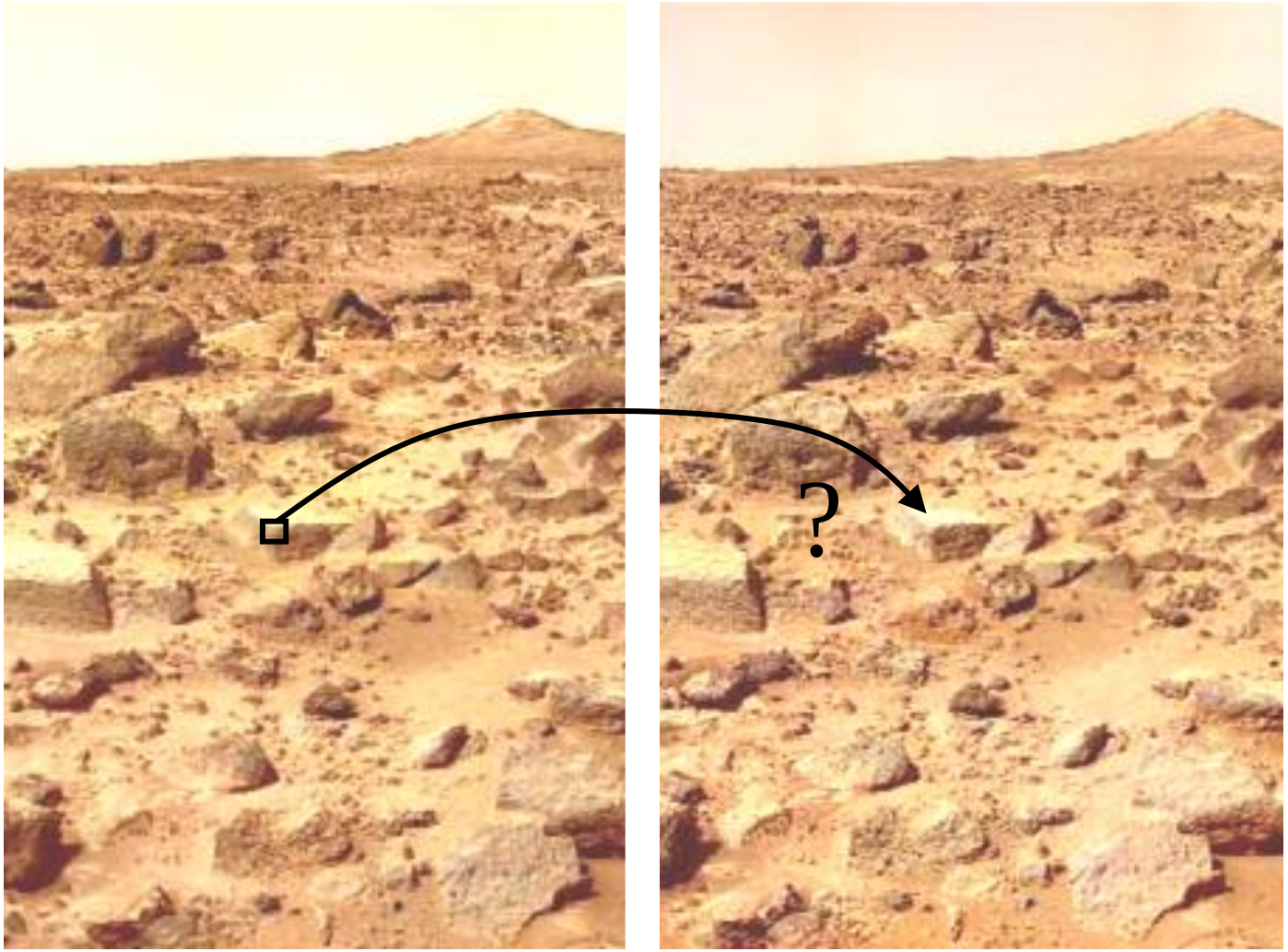
Establishing correspondence



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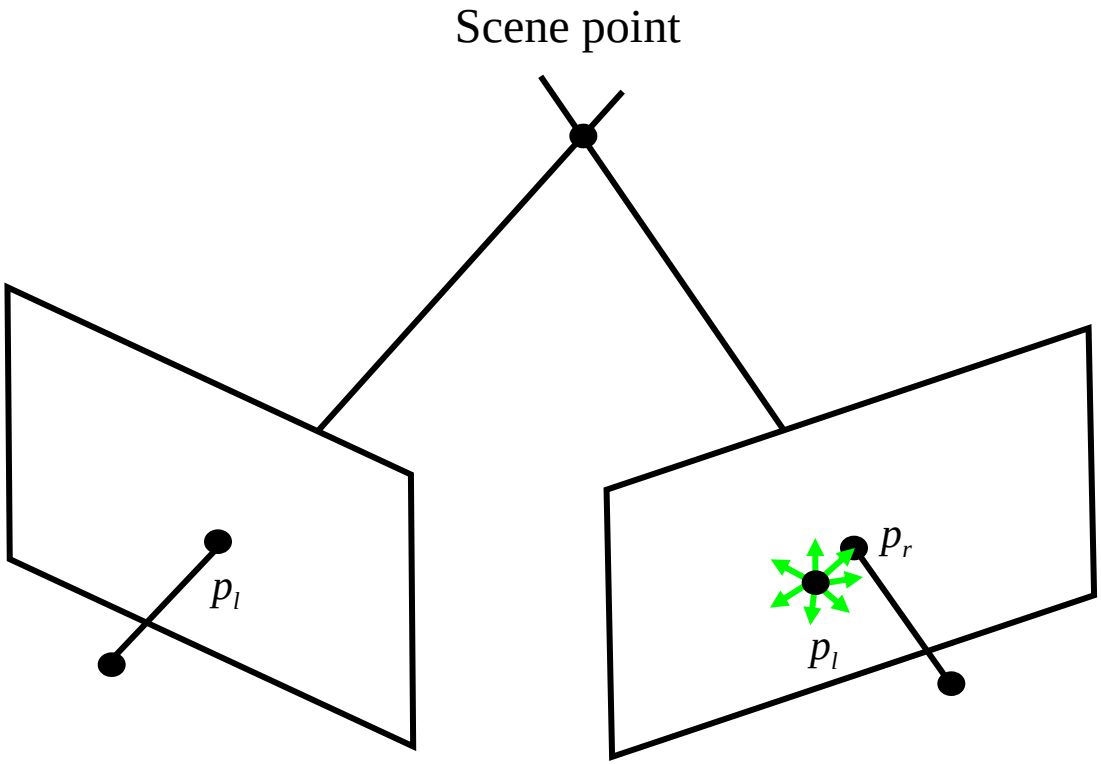
Epipolar geometry



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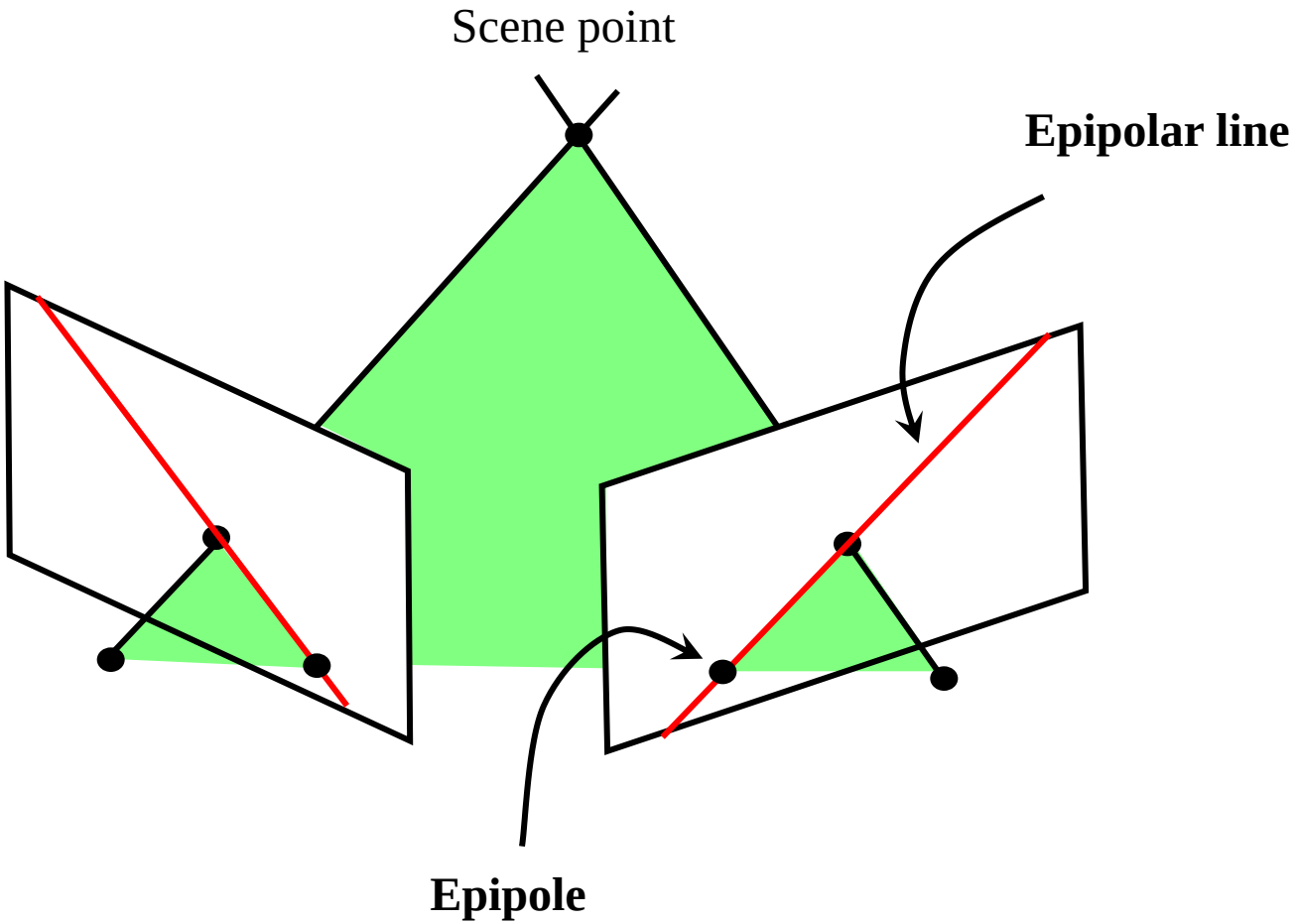
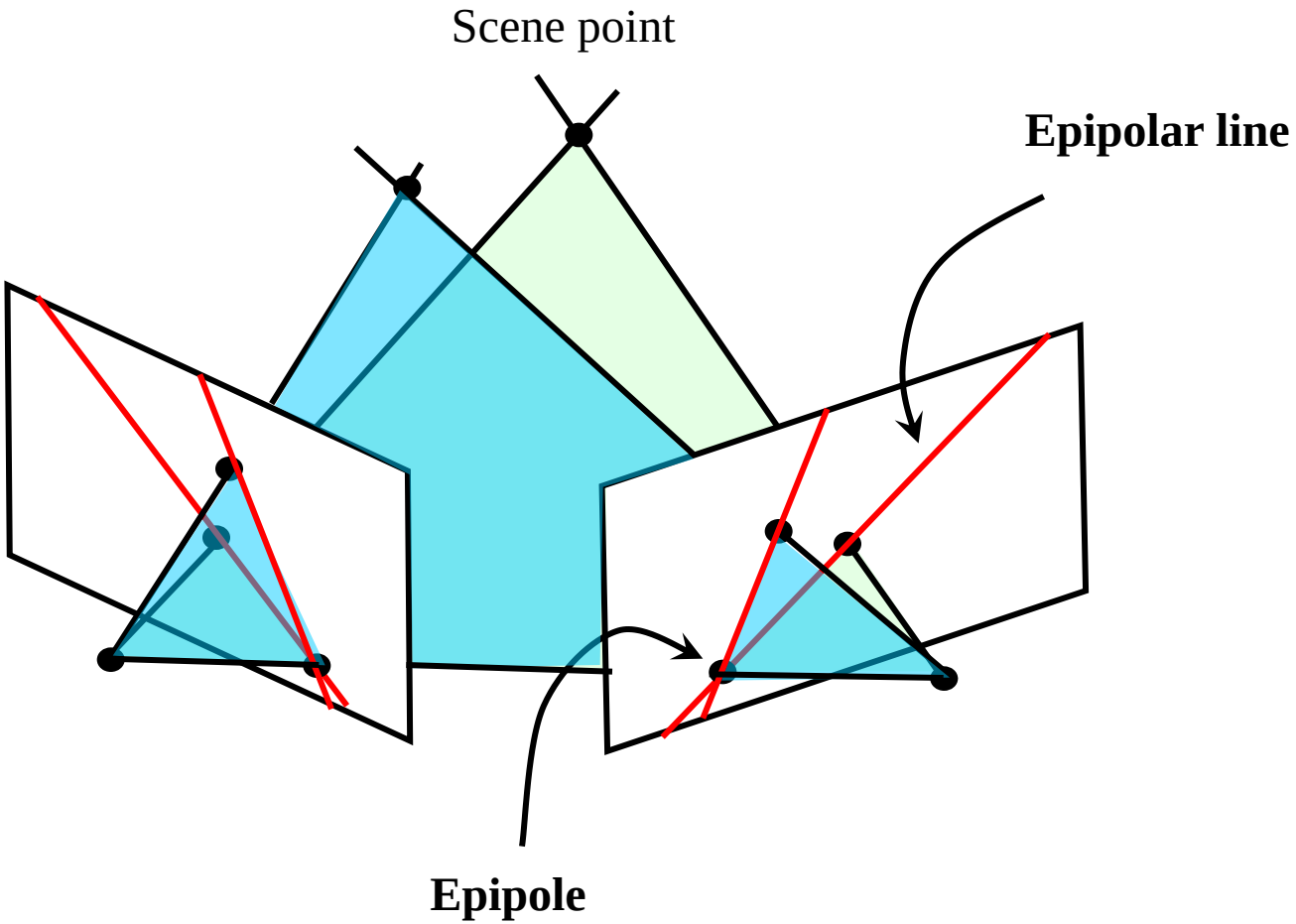
Epipolar geometry



Stereopsis

Epipolar geometry



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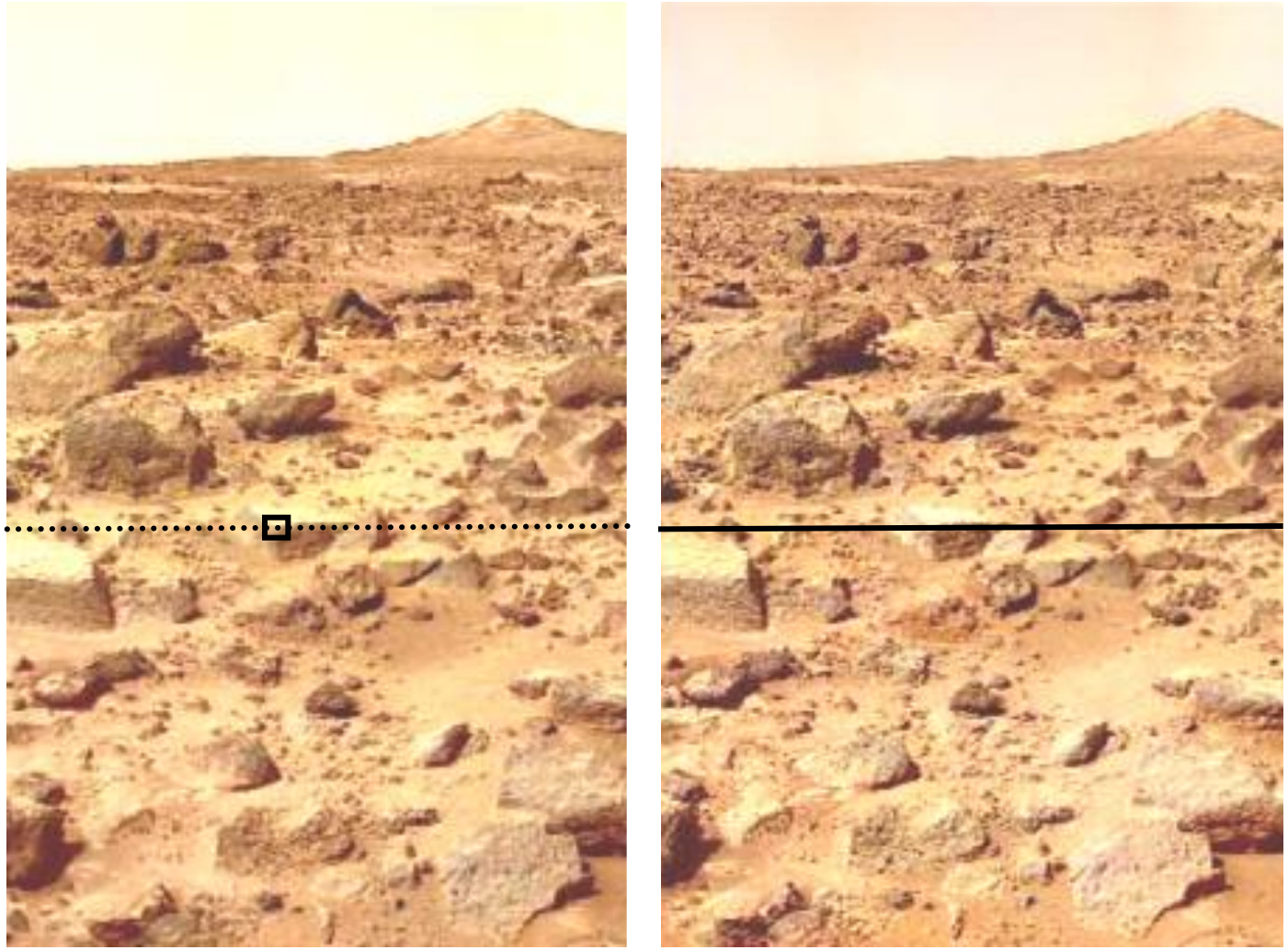
Rectified epipolar geometry



Stereopsis

Intensity-based methods for establishing correspondence

Assume:

Intensity of surfaces is independent of viewing angle

Establish correspondence for pixel *p* in the left image:

Define a window W_p of size $(2n+1)$x$(2n+1)$ around p

For each pixel q along p's epipolar line in the right image do

Define a window W_q of size $(2n+1)$x$(2n+1)$ around q

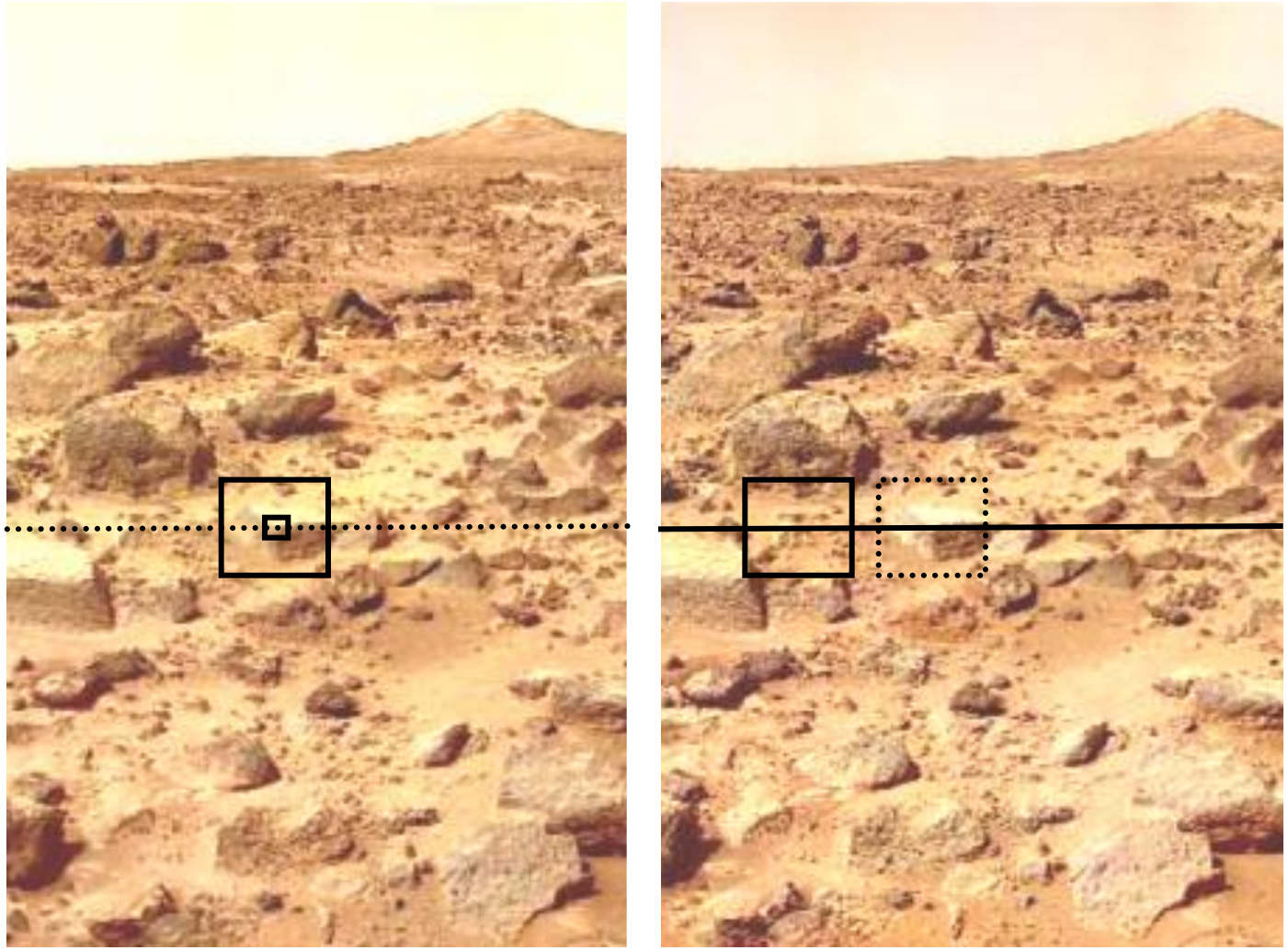
Compute similarity $S(W_p,W_q)$ beween W_p and W_q

End

Choose q which maximizes $S(W_p,W_q)$

Stereopsis

Intensity-based methods for establishing correspondence



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Intensity based similarity measures

$$SSD(W_p, W_q) = \sum_{\vec{w} \in W} \left[I_l(\vec{p} - \vec{w}) - I_r(\vec{q} - \vec{w}) \right]^2 \Rightarrow \min$$

$$Cor(W_p, W_q) = \sum_{\vec{w} \in W} I_l(\vec{p} - \vec{w}) \cdot I_r(\vec{q} - \vec{w}) = \vec{W}_p \cdot \vec{W}_q \Rightarrow \max$$

$$NC(W_p, W_q) = \frac{\left(\vec{W}_p - \overline{W}_p\right) \cdot \left(\vec{W}_q - \overline{W}_q\right)}{\left|\vec{W}_p - \overline{W}_p\right| \cdot \left|\vec{W}_q - \overline{W}_q\right|} \Rightarrow \max$$

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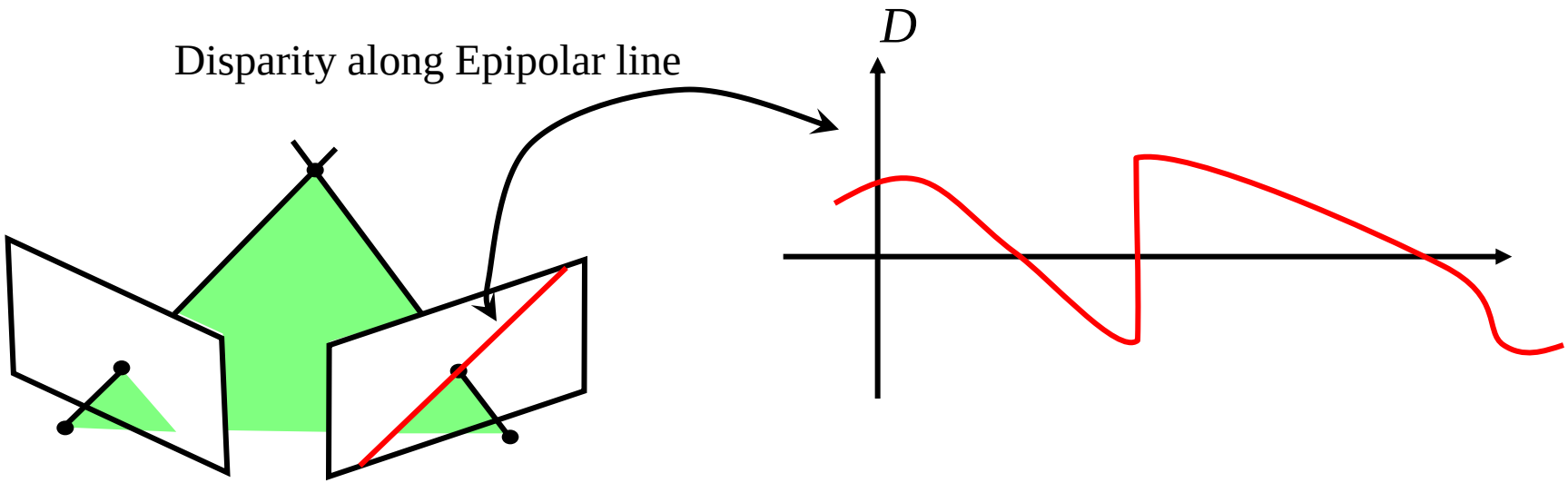
Correspondence via global minimization



$$E_s[D] = \int_{x'}\int_{y'} smoothness[D(x',y')]dx'dy'$$

$$E_i[D] = \int_{x'}\int_{y'} similarity[I_l(x',y'), I_r(x'+D(x',y'),y')]dx'dy'$$

Stereopsis

Correspondence via global minimization

Common penalty functions:

$$E_s = \int_{x'}\int_{y'} \|\nabla D\|^2 \, dx' \, dy' \qquad E_s = \int_{x'}\int_{y'} \left(\nabla^2 D\right)^2 dx' \, dy'$$

$$E_i = \int_{x'}\int_{y'} \left[I_l(x', y') - I_r(x' + D(x', y'), y')\right]^2 dx' \, dy'$$

Establishing correspondence via minimization:

$$\arg\min_D E_s[D] + \lambda E_i[D]$$

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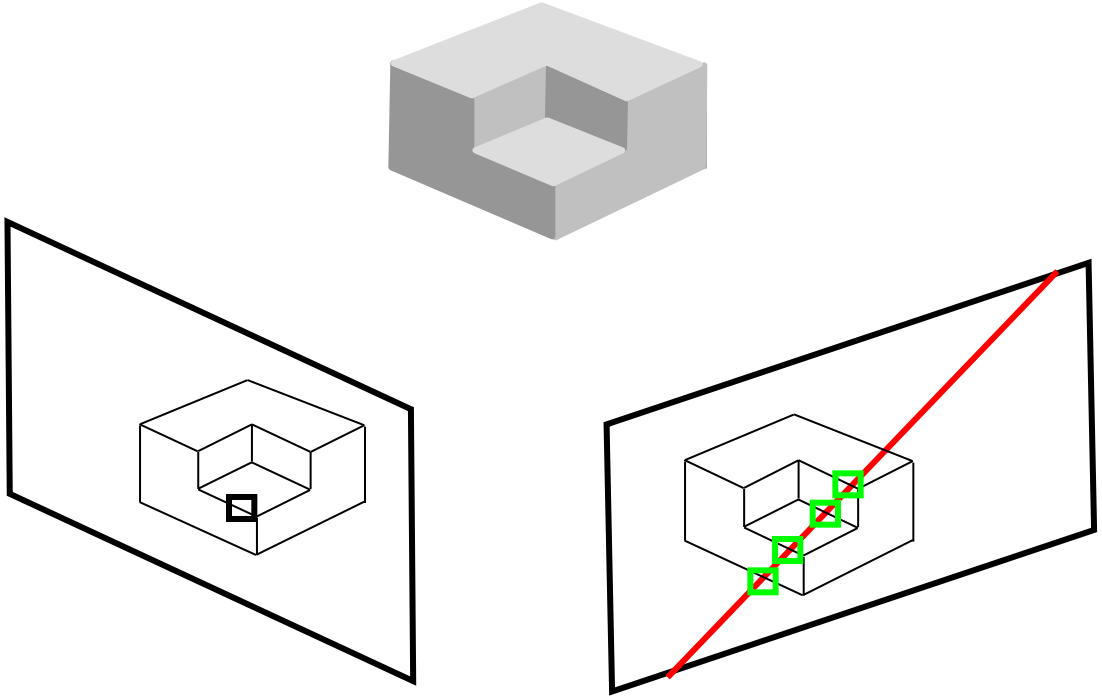
Feature-based methods for establishing correspondence



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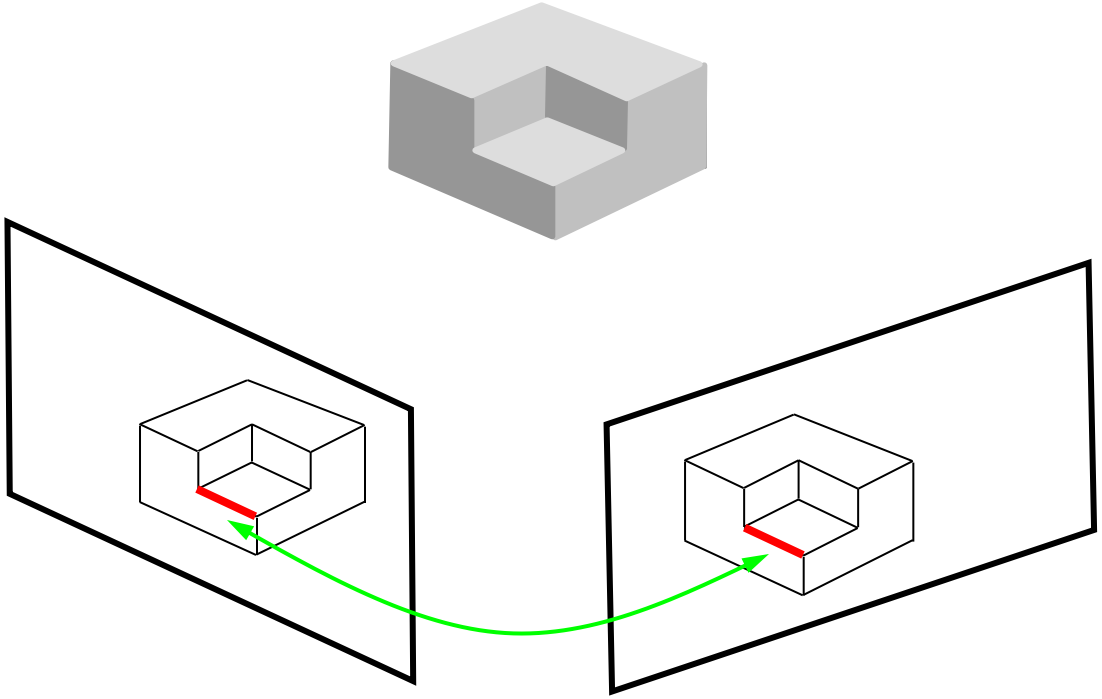
Feature-based methods for establishing correspondence



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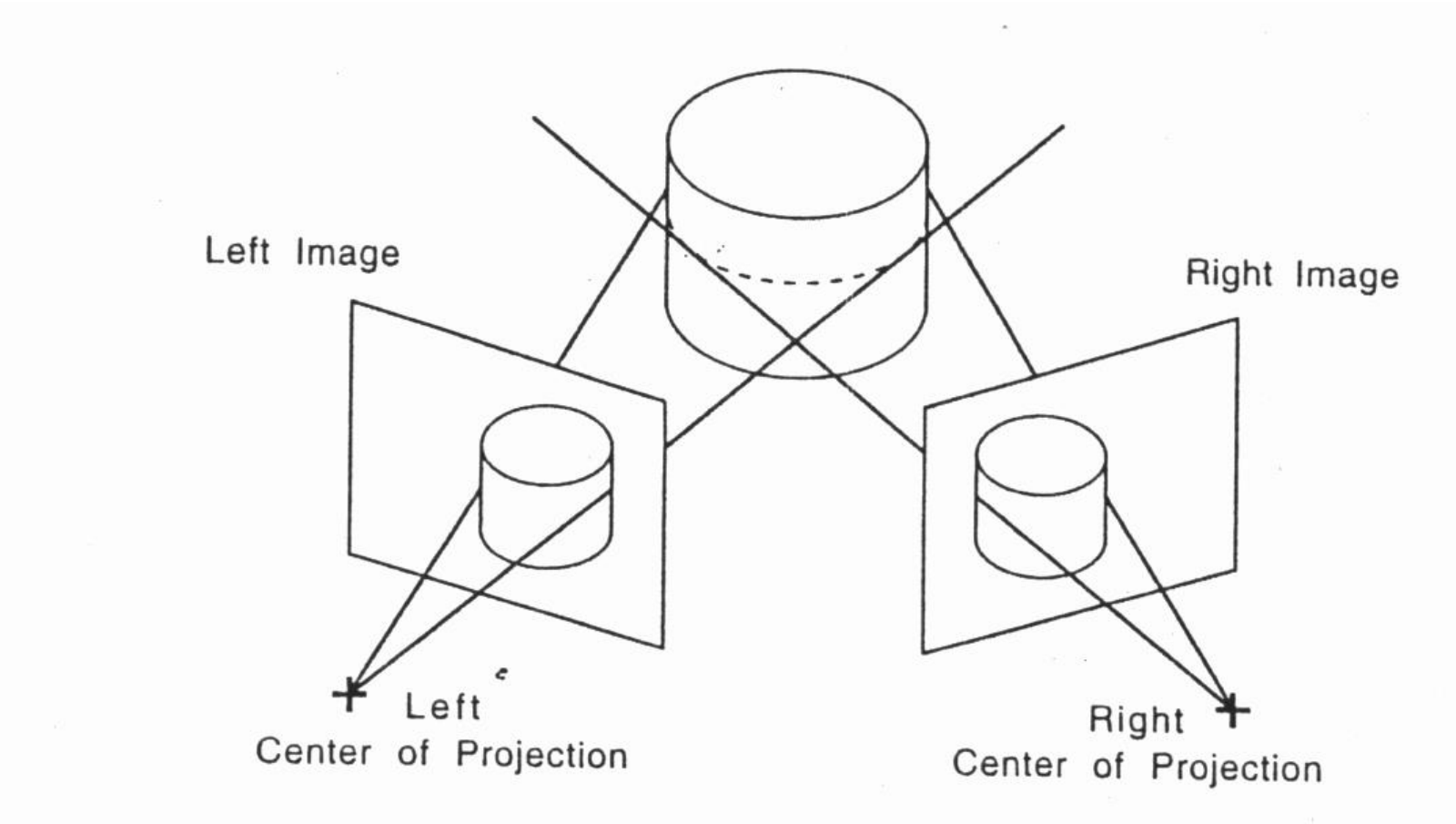
Feature-based methods for establishing correspondence